STAT-S 650 HOMEWORK 4

MANISH
SURESH MANDLALYA

Theory

1). Given $Y(t) = \alpha \cdot Y(t-1) + W(t) + \beta \cdot W(t-1)$

Given spectral density of ARMA from textbook.

$$f_x(w) = \frac{\sigma_w^2 \left| \theta(e^{-2\pi i w}) \right|_2}{\left| \phi(e^{-2\pi i w}) \right|^2}$$

Where $\phi(z) = 1 - \sum_{k=1}^{p} \phi_k z^k$

$$\theta(z) = 1 + \sum_{k=1}^{q} \theta_k z^k$$

$W(t)$ is white noise.

-> lets assume $BY(t) = Y(t-1)$

$B W(t) = W(t-1)$.

$$Y(t) - \alpha B Y(t) = (1 + \beta B) W(t)$$

$$\Rightarrow (1 - \alpha B) Y(t) = (1 + \beta B) W(t).$$

$\Rightarrow$ Transfer function is $H(B) = \frac{1 + \beta B}{1 - \alpha B}$

Using the property

$$f_Y(\omega) = \sigma_w^2 \frac{\theta |e^{-2\pi i \omega}|^2}{\phi |e^{-2\pi i \omega}|^2}$$

$$\Rightarrow \theta(e^{-2\pi i \omega}) = 1 + \beta e^{-2\pi i \omega}$$
$$\phi(e^{-2\pi i \omega}) = 1 - \alpha e^{-2\pi i \omega}.$$

$$f_Y(\omega) = \sigma_w^2 \frac{|1+\beta e^{-2\pi i \omega}|^2}{|1-\alpha e^{-2\pi i \omega}|^2}$$

$$\Rightarrow |1+\beta e^{-2\pi i \omega}|^2 = (1+\beta e^{-2\pi i \omega})(1+\beta e^{2\pi i \omega})$$
$$= 1 + \beta^2 + 2\beta \cos(2\pi\omega)$$

$$|1-\alpha e^{-2\pi i \omega}|^2 = (1-\alpha e^{-2\pi i \omega})(1-\alpha e^{2\pi i \omega}) = 1 + \alpha^2 - 2\alpha\cos(2\pi\omega)$$

∴ Final spectral density

$$f_Y(\omega) = \sigma_w^2 \frac{1 + \beta^2 + 2\beta\cos(2\pi\omega)}{1 + \alpha^2 - 2\alpha\cos(2\pi\omega)}$$

Hence proved.

2) AR (2) process $w(t)$ is noise

given

$$y(t) = \alpha_1 \cdot y(t-1) + \alpha_2 \cdot y(t-2) + w(t)$$

→ lets assume L is the lag operator

$$Ly(t) = y(t-1)$$

AR(2) model becomes.

$$(1 - \alpha_1 L - \alpha_2 L^2)\, y(t) = w(t)$$

→ $\phi(L) = 1 - \alpha_1 L - \alpha_2 L^2$.

$$y(t) = \frac{1}{\phi(L)}\, w(t).$$

Replacing L with $e^{-i\omega}$

$$\phi(e^{-i\omega}) = 1 - \alpha_1 e^{-i\omega} - \alpha_2 e^{-i2\omega}.$$

→ Power Spectrum

$$S_y(\omega) = \frac{\sigma_w^2}{|\phi(e^{-i\omega})|^2} = \frac{\sigma_w^2}{|1 - \alpha_1 e^{-i\omega} - \alpha_2 e^{-i2\omega}|^2}$$

$$|\phi(e^{-i\omega})|^2 = \phi(e^{i\omega}) \cdot \phi(e^{i\omega}).$$

$$= 1 + \alpha_1^2 + \alpha_2^2 - 2\alpha_1(1+\alpha_2)\cos\omega - 2\alpha_2 \cos(2\omega)$$

∴ Final Spectrum of AR(2) → $\dfrac{\sigma_w^2}{1 + \alpha_1^2 + \alpha_2^2 - 2\alpha_1(1+\alpha_2)\cos\omega - 2\alpha_2 \cos(2\omega)}$